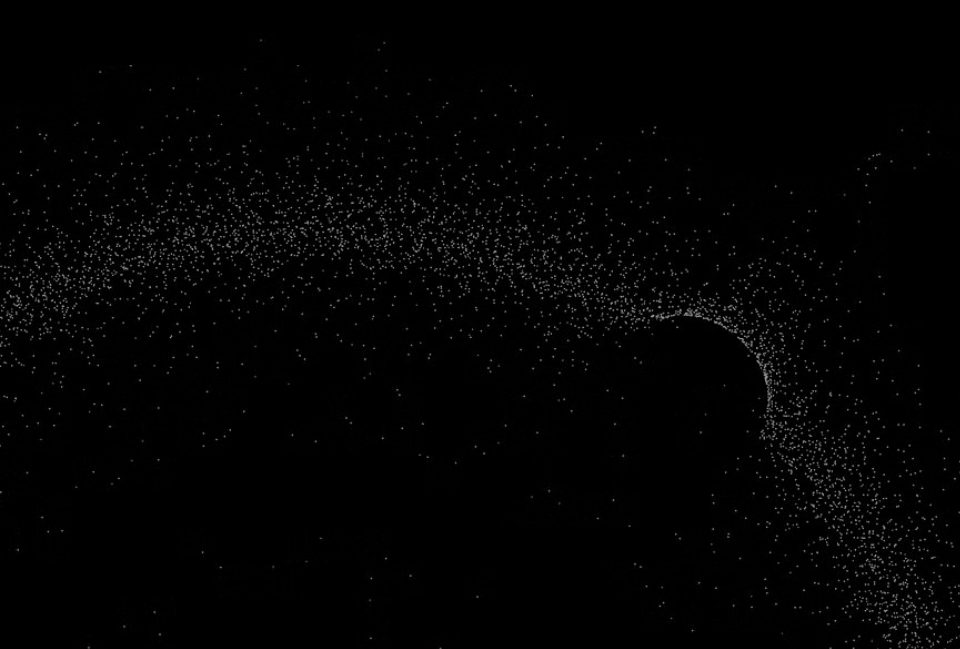


Working with the BMDC

Ali Kahlert, MS

Agenda



1 Introduction

2 BMDC Capabilities

3 Project Kick-Off Meeting

4 Later Steps

5 Random Tips

6 Questions

Introduction

Education

- BS (2015) Mortuary Science
 - Still licensed; non-practicing
- MS (2023) Human Factors & Ergonomics
- PhD (TBD) Human Factors & Ergonomics

Current Role

- Lab Supervisor
- Design & Prototyping Support
- Model Builder
 - Digital
 - Analog



BMDC Capabilities: MedWorx

3D Printing

- FDM, Resin, and Polyjet printers
- Different cost structures
- FDM printers self-operable (except one)
- Resin printers self-operable, but require you to purchase materials

Digital Models

- Segmentation
 - Materialise Mimics
- Clean-Up
 - Blender
 - Materialize 3Matic
 - Meshmixer

Casting (NEW)

- Casting, small amounts of manufacturing
 - 100 items (yes, probably)
 - 10,000 items (nope)
- Other materials

Digital Fabrication

- Augmentation of files
- Added pathologies, injuries, display supports, color-coding, etc.

More Capabilities

VR

- Any digital model can be put into VR
- Use Unity Engine

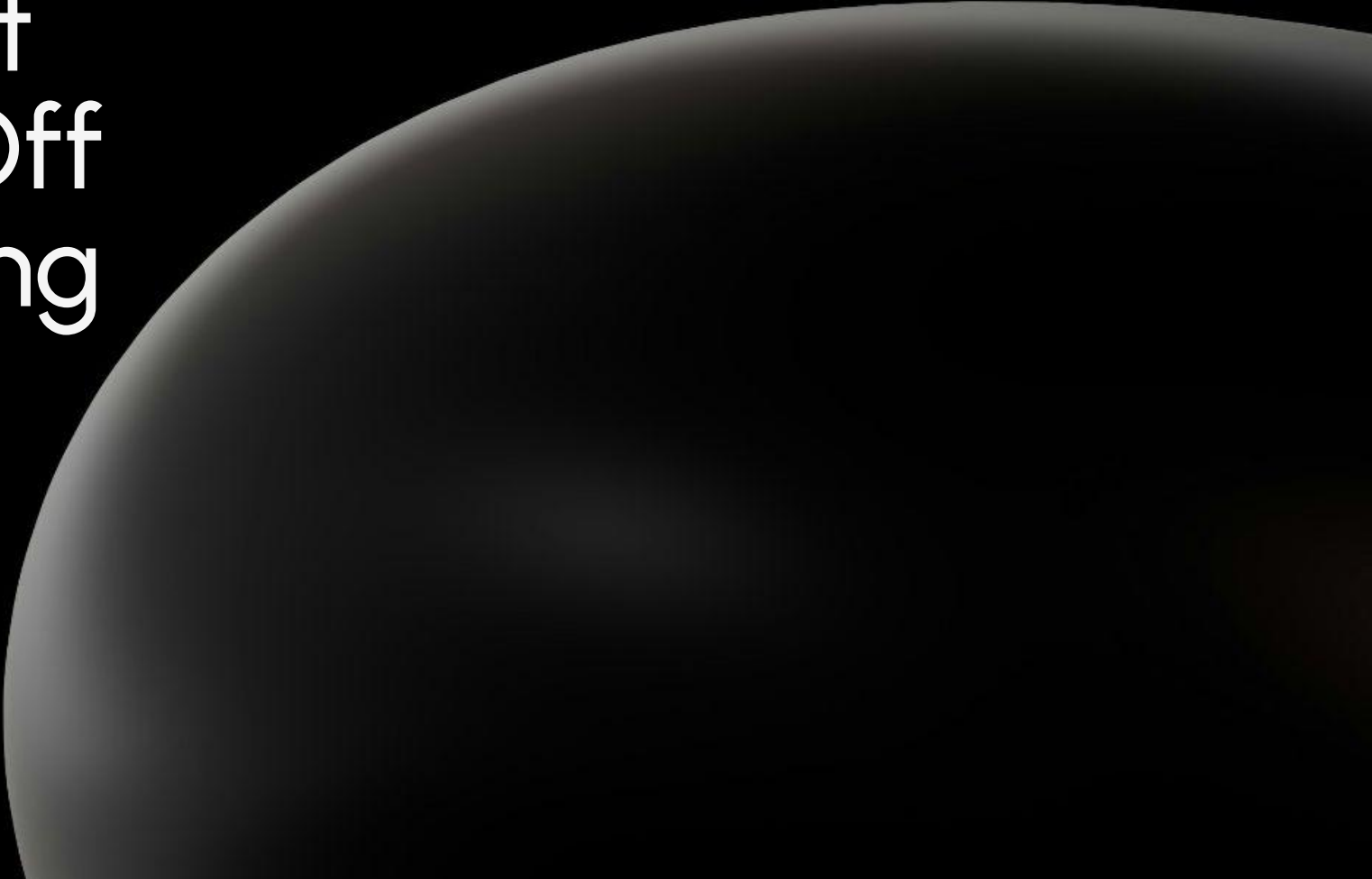
Machining

- CNC Machine
- Lathe
- Laser Cutter
- Drill Press
- Belt Sander

Public Computers

- SolidWorks
- Prusa Slicer
- Other software

Project Kick-Off Meeting



What to Know

Your Project

- Know what you are trying to do
- We may challenge you
- We can't do everything

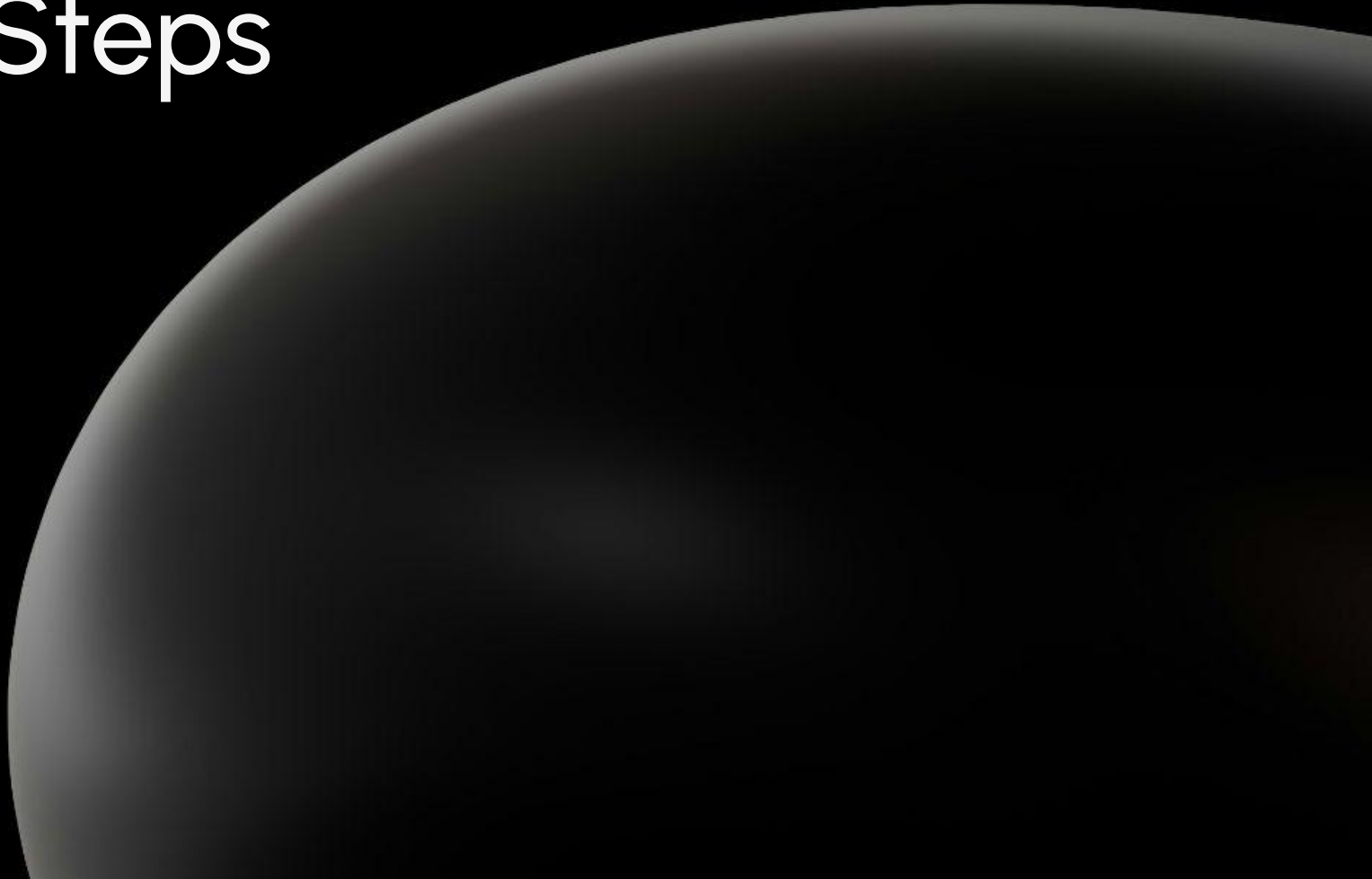
Your Budget

- Regrettably, we aren't a free service :(
- We cannot accept personal credit cards
- Use departmental chart strings, or a business we can invoice
- Finding funds
- Work within limited budgets

Your [Desired] Deliverables

- What do you want at the end of this project?
- Staggering deliverables
- Moving goalposts moves deadlines back

Later Steps



Later Steps

Ongoing Feedback

- Really hard to move forward if you cannot be reached for feedback
- Helpful if you have 3D viewing software
 - Use free stuff

Timely Delivery of Needed Items

- Patient scans
- Measurements
- Documentation
- If the project hinges on these things... we need them sooner rather than later

IRB & Beyond

- If you are doing a study, you handle the IRB
- We can assist you if you want to patent/license/take product to market
 - Connect you to the right people

Random Tips



Random Tips

- MRI vs. CT
 - We prefer CTs
 - MRIs are tricky
- All files require some repair/cleanup
 - It is worth it; I promise
- Grad students (\$\$\$) vs Undergrads (\$)
 - Complex files require more work
 - More complex work falls to the grad students
- Expect to pay several hundred dollars for one polyjet print
 - Can easily be several thousand
 - Our record: \$18k
- Build in as much time/money as you can
 - Unforeseen things happen
 - Students get sick, printers break

These tips are meant to level set expectations and make our working relationship as seamless as possible!

We want to help you!

Questions

